STA 419 — MEASURE AND PROBABILITY THEORY

Chapter 1 — Sets, Fields, Measures, Expectation, Modes of Convergence and Limit Theorems

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This edition contains the whole of the Chapter 1 notes (33 pages in the source file, every page accounted for in the source map at the end). The prose and the mathematics are those of the notes; the working has been re-set in proper mathematical form and the few places where the source's algebra is mistyped or incomplete are corrected, with each correction listed in the appendix.

On the "overlaid" appearance of the downloaded page: the saved web page stores every page **twice** — once as a background picture of the page and once again as an invisible layer of absolutely positioned text — and the two layers are drawn on top of each other. When the file is opened or printed, the picture and the text layer are both painted, so the same sentence appears twice, slightly displaced, and the pages (which also sit inside a scaled scrolling container) collide with one another. Printing the page as it stands produces 34 cramped pages of doubled text. The picture layer is moreover incomplete: in the original document several glyphs — the ligatures *fi*, *ff*, the script letters of $\mathcal{F}$ and $\mathcal{B}$ and the symbol $\mathbb{R}$ — were drawn in fonts that were never embedded, so they are missing from the pictures, while the text layer preserves them. This edition is therefore built from the text layer alone, so that each page carries one clean copy of its content.*

Notation used throughout. $(U, \mathcal{F}, P)$ is a probability space: U is the universal (sample) set, $\mathcal{F}$ the σ-field of events and P the probability measure. $\mathcal{B}$ is the Borel σ-field of the real line, $\mathbb{R}$ the real line, $\varnothing$ the empty set, A^c the complement of A in U, and I_A the indicator function of A.

1.1 DEFINITIONS

Set. A set is a collection or group of objects with similar characteristics.

Null (empty) set. A set with no objects inside it; denoted $\varnothing$.

Examples: the sets

$$A = \{a, b, c\}, \qquad B = \{a, e\}, \qquad C = \{d\}, \qquad \text{null set} = \varnothing = \{\}$$

Universal set. The set containing all possible objects. In the examples above the universal set is

$$U = \{a, b, c, d, e\}.$$

Class. A set whose objects are sets. For example, the set

$$\tau = \{A, B, C\}, \qquad \alpha = \{D, E\}$$

is a class whose elements are the sets A, B, C. The distinction in the notes is

Level	Objects
elements	$a,\ b,\ c,\ d,\ e,\ f$
sets	$A,\ B,\ C,\ D,\ E,\ F$
classes	$\tau,\ \alpha$

Subset. Let A be a set and let B be another set. Then A is a subset of B if all the elements of A are members of B, that is

$$A \subset B \quad \Leftrightarrow \quad \left(w \in A \Rightarrow w \in B \right).$$

Equality. If all members of A are members of B and all members of B are members of A, then A and B are equivalent, that is $A = B$; equivalently $A \subset B$ and $B \subset A$.

Superset. If A is a subset of B, then B is a superset of A:

$$A \subset B \ \Rightarrow \ B \supset A.$$

Disjoint sets. Two sets are disjoint if they have no elements in common, that is $A \cap B = \varnothing$.

Union. The union of two or more sets is the set containing all possible elements of all the sets, e.g.

$$A \cup B = \{a, b, c, e\}.$$

Intersection. The intersection of two sets A and B is the set of all elements common to A and B; written $A \cap B$.

Difference. The difference $A - B$ is the set of all elements that are in A but not in B:

$$A - B = \{w: w \in A,\ w \notin B\} = A \cap B^c.$$

Complement. Let A be a subset of the universal set U. The complement of A, written A^c, is the set of all elements of U that are not in A:

$$A^c = \{w: w \in U,\ w \notin A\} = U - A.$$

Power set. Let A be a set. The power set of A is the class consisting of all possible subsets of A. For $A = \{a, b, c\}$ the subsets are

Number of elements	Subsets
0	$\varnothing$
1	$\{a\},\ \{b\},\ \{c\}$
2	$\{a, b\},\ \{a, c\},\ \{b, c\}$
3	$\{a, b, c\}$

the last of these showing that a set is always a subset of itself. Hence

$$\mathcal{P}(A) = \big\{\varnothing,\ \{a\},\ \{b\},\ \{c\},\ \{a, b\},\ \{a, c\},\ \{b, c\},\ \{a, b, c\}\big\}.$$

1.2 SEQUENCES AND LIMITS

A sequence of sets $A_1, A_2, \ldots, A_n, \ldots$ is denoted by $\{A_n,\ n = 1, 2, \ldots\}$ or simply by $\{A_n\}$.

Monotone increasing. A sequence is monotone increasing if

$$A_n \subseteq A_{n+1} \qquad \text{for each } n.$$

For such a sequence the partial unions satisfy

$$\bigcup_{i=1}^{n} A_i = A_n,$$

and the sequence is denoted by $A_n \uparrow A$. Its limit is the union of all the terms:

$$\lim_{n \to \infty} A_n = \bigcup_{n=1}^{\infty} A_n = A.$$

Monotone decreasing. If on the other hand A_n is a superset of A_{n+1}, that is

$$A_n \supseteq A_{n+1} \qquad \text{for every } n,$$

then the sequence is monotone decreasing. For such a sequence

$$\bigcap_{i=1}^{n} A_i = A_n,$$

and the limit is the intersection of all the terms:

$$\lim_{n \to \infty} A_n = \bigcap_{n=1}^{\infty} A_n = A, \qquad \text{written } A_n \downarrow A.$$

Example 1. Let

$$A_n = \left\{w: 2 - \frac{1}{n} < w < 7 - \frac{2}{n}\right\}.$$

Then

$$A_1 = \{w: 1 < w < 5\}, \qquad A_2 = \{w: 1.5 < w < 6\}, \qquad A_3 = \{w: 1.67 < w < 6.33\},$$

an increasing sequence, and the limit is

$$A = \{w: 2 \le w < 7\}.$$

> *[The notes give the limit as $\{w: 2 < w \le 7\}$. With the inequality "$2 - \frac{1}{n} < w$" the point $w = 2$ is a member of every A_n and is therefore in the limit, while with "$w < 7 - \frac{2}{n}$" the point $w = 7$ belongs to no A_n and is therefore excluded; the limit is $\{w: 2 \le w < 7\}$.]*

Example 2. Let

$$B_n = \left\{ w: 2 + \frac{3}{n} < w < 6 - \frac{1}{n} \right\}.$$

Then

$$B_1 = \{w: 5 < w < 5\} = \varnothing, \qquad B_2 = \{w: 3.5 < w < 5.5\}, \qquad B_3 = \{w: 3 < w < 5.67\},$$

an increasing sequence with limit

$$B = \{w: 2 < w < 6\}.$$

1.3 FIELDS

A **field** is a non-empty class $\mathcal{F}$ of subsets of U which is

1. closed under complementation, and
2. closed under finite intersections.

Let $\mathcal{F}$ be a field. Then

1. if $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$;
2. if $A \in \mathcal{F}$ and $B \in \mathcal{F}$ then $A \cap B \in \mathcal{F}$.

1.3.1 DeMorgan's rule

It states that if $A, B \in \mathcal{F}$ then $A \cup B \in \mathcal{F}$.

Proof. Let $A, B \in \mathcal{F}$. By rule 1,

$$A^c \in \mathcal{F}, \qquad B^c \in \mathcal{F},$$

and by rule 2 applied to A^c and B^c,

$$\left(A^c \cap B^c\right) \in \mathcal{F},$$

hence by rule 1 again

$$\left(A^c \cap B^c\right)^c \in \mathcal{F}.$$

But by DeMorgan's law $\left(A^c \cap B^c\right)^c = A \cup B$, so that

$$A \cup B \in \mathcal{F}. \qquad \blacksquare$$

Result 1. The empty set $\varnothing$ is a member of every field.

Proof. If $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$ by rule 1, so by rule 2

$$A \cap A^c = \varnothing \in \mathcal{F}. \qquad \blacksquare$$

Result 2. The universal set U is a member of every field.

Proof. If $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$ by rule 1, so by DeMorgan's rule

$$A \cup A^c = U \in \mathcal{F}. \qquad \blacksquare$$

Minimal field. Let A be a set and U the universal set. The minimal field of the set A is the smallest field containing A, namely

$$\mathcal{F}_A = \left\{\varnothing,\ A,\ A^c,\ U\right\}.$$

This is the field said to be *generated* by A.

σ-field. A σ-field is a field which is

1. closed under complementation, and
2. closed under countable intersections and countable unions.

Borel field. A Borel field is a field which contains the intervals of the real line:

$$(-\infty, a), \quad (-\infty, a], \quad (b, \infty), \quad [b, \infty), \quad (a, b), \quad [a, b), \quad (a, b], \quad [a, b],$$

and is denoted by $\mathcal{B}$. It is the σ-field generated by the class of all such intervals.

Monotone field. A field $\mathcal{F}$ is a monotone field if it is closed under monotone operations: if $\{A_n\}$ is a monotone sequence in $\mathcal{F}$ and $\lim A_n \in \mathcal{F}$, then $\mathcal{F}$ is a monotone field.

Theorem. A σ-field is a monotone field, and conversely a monotone field is a σ-field.

Proof. Let $\mathcal{F}$ be a σ-field and let $\{A_n\}$ be a sequence in $\mathcal{F}$. If $A_n \uparrow A$, then

$$A = \bigcup_{i=1}^{\infty} A_i$$

is a countable union of sets of $\mathcal{F}$, and $A \in \mathcal{F}$ since $\mathcal{F}$ is a σ-field. Similarly, if $A_n \downarrow A$, then

$$A = \bigcap_{n=1}^{\infty} A_n$$

is a countable intersection of sets of $\mathcal{F}$, and $A \in \mathcal{F}$ since $\mathcal{F}$ is a σ-field. These are monotone operations, so $\mathcal{F}$ is a monotone field.

Conversely, let $\mathcal{F}$ be a monotone field and let $A_1, A_2, \ldots, A_n, \ldots \in \mathcal{F}$. Put

$$A' = \bigcup_{i=1}^{\infty} A_i, \qquad A'' = \bigcap_{i=1}^{\infty} A_i.$$

The partial unions $C_n = \bigcup_{i=1}^{n} A_i$ form an increasing sequence in $\mathcal{F}$ with limit A', and the partial intersections $D_n = \bigcap_{i=1}^{n} A_i$ form a decreasing sequence in $\mathcal{F}$ with limit A''. Since $\mathcal{F}$ is closed under monotone operations, $A' \in \mathcal{F}$ and $A'' \in \mathcal{F}$. Hence countable unions and countable intersections of members of $\mathcal{F}$ belong to $\mathcal{F}$, and $\mathcal{F}$ is a σ-field. ■

1.3.2 FUNCTIONS

Let U be a universal set and let $w \in U$. Let $X(w) = w'$ be the value associated with w when w is operated on by X. If $w' \in U'$ for every $w \in U$, then X is a function, or mapping, that maps U into U', written

$$X: U \to U'.$$

U is the domain and U' the range (codomain) of X. For example:

1. **one-to-one** functions — each w gives a different w';
2. **one-to-many** — one w is associated with several values;
3. **many-to-one** — several w give the same w'.

Real-valued functions. These are functions which map into the real line, $X: U \to \mathbb{R}$.

Set functions. These are functions whose domain contains sets, that is, they operate on sets.

Inverse functions. An inverse function is a set function. Let $X: U \to U'$ and let $D \subset U'$. Then

$$X^{-1}(D) = \left\{ w: X(w) = w' \in D \right\} \subset U.$$

Measurable functions. Let U be a universal set and let $\mathcal{F}$ be a σ-field over U, so that all sets of $\mathcal{F}$ are subsets of U. Let X be a mapping of U into the real line, $X: U \to \mathbb{R}$, and let $B \in \mathcal{B}$, where $\mathcal{B}$ is the Borel field. If

$$X^{-1}(B) = A, \qquad \text{where } A \in \mathcal{F},$$

then X is a measurable function with respect to the σ-field $\mathcal{F}$.

Borel functions. A function is a Borel function if it maps the real line into the real line and is measurable with respect to $\mathcal{B}$.

Measurable space. Let $X: U \to \mathbb{R}$ be such that for every $B \in \mathcal{B}$ the set $X^{-1}(B) = A$ lies in $\mathcal{F}$, $\mathcal{F}$ being a σ-field over U. Then the pair $(U, \mathcal{F})$ is called a measurable space.

Measure. Let $X: U \to \mathbb{R}$ and let $(U, \mathcal{F})$ be a measurable space. Define a function μ on $\mathcal{F}$ such that

1. $\mu(\varnothing) = 0$;
2. $\mu(A) \geq 0$ for all $A \in \mathcal{F}$;
3. μ is countably additive: if $A_1, A_2, \ldots, A_n, \ldots$ are pairwise disjoint sets of $\mathcal{F}$, then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu\left(A_i\right).$$

Then μ is called a **measure** on the measurable space $(U, \mathcal{F})$, and the triple $(U, \mathcal{F}, \mu)$ is a **measure space**.

If $\mu(U) < \infty$ then μ is a **finite measure**. If $\mu(U) = \infty$ but there exist $A_1, A_2, \ldots$ with $\bigcup_{i=1}^{\infty} A_i = U$ and $\mu(A_i) < \infty$ for each i, then μ is a **σ-finite measure**.

Indicator functions. Let U be a universal set and let $A \subset U$. The indicator function of the set A is defined by

$$I_A(w) = \begin{cases} 1, & w \in A, \\ 0, & w \in A^c. \end{cases} \qquad (1.1)$$

Simple functions. A simple function is a finite linear combination of indicator functions of sets which are not necessarily disjoint:

$$X = C_A I_A + C_B I_B, \qquad (1.2)$$

where C_A and C_B are constants.

Theorem. Any simple function X can be written in the form

$$X = \sum_{i=1}^{n} x_i I_{A_i},$$

where the sets A_i are pairwise disjoint.

Proof. Let B_1 and B_2 be subsets of U and let

$$X = C_1 I_{B_1} + C_2 I_{B_2}.$$

For $w \in U$ there are four possibilities:

Case	Value	Set on which it occurs
$w \notin B_1,\ w \notin B_2$	$X(w) = 0 = x_1$	$A_1 = B_1^c \cap B_2^c$
$w \in B_1,\ w \notin B_2$	$X(w) = C_1 = x_2$	$A_2 = B_1 \cap B_2^c$
$w \notin B_1,\ w \in B_2$	$X(w) = C_2 = x_3$	$A_3 = B_1^c \cap B_2$
$w \in B_1,\ w \in B_2$	$X(w) = C_1 + C_2 = x_4$	$A_4 = B_1 \cap B_2$

The sets A_1, A_2, A_3, A_4 are pairwise disjoint and their union is U; hence

$$X = \sum_{i=1}^{4} x_i I_{A_i}. \qquad \blacksquare$$

Partition. Let U be a universal set and let $A_1, A_2, \ldots, A_n, \ldots$ be non-empty, pairwise disjoint subsets of U such that

$$\bigcup_i A_i = U.$$

Then the class $\{A_i;\ i = 1, 2, \ldots, n\}$ is a partition of U.

1.3.3 PROBABILITY

Probability is a numerical measure of the chance of occurrence of events. Let A be an event. Then

1. $0 \le P(A) \le 1$;
2. $P(\varnothing) = 0$;
3. if $\{A_i;\ i = 1, 2, \ldots, n\}$ is a partition of U, then

$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P\left(A_i\right).$$

> *[The notes print item 2 as $p(0) = 1$; the probability of the impossible event is $P(\varnothing) = 0$.]*

Definitions.

- *Experiment* — an activity whose outcome cannot be determined with certainty.
- *Sample space* — the set of all possible outcomes.
- *Outcome* — an element of the sample space.

For example, tossing a coin is an experiment and the outcomes are H and T:

$$S = \{H, T\}, \qquad E_1 = \{\varnothing\}, \qquad E_2 = \{H\}, \qquad E_3 = \{T\}, \qquad E_4 = \{H, T\}.$$

1.4 APPROACHES TO PROBABILITY

1. Classical concept

All outcomes are assumed to be equally likely. The probability of an event is the number of outcomes favouring that event divided by the total number of outcomes in the sample space. For two tosses of a coin,

$$S = \{HH, HT, TH, TT\},$$

and the event of getting a head first or no head is $E = \{HH, HT, TT\}$; therefore

$$P(E) = \frac{3}{4}.$$

> *[The notes print $S = \{HH, HT, TH, HH\}$; the fourth outcome is TT.]*

2. Statistical (limiting) concept

Suppose an experiment is conducted n times, and assume there are k different possible events, that is k different sets of outcomes. Let the frequency distribution be

Event	Frequency	Relative frequency
E_1	f_1	r_1
E_2	f_2	r_2
E_i	f_i	r_i
⋮	⋮	⋮
E_k	f_k	r_k

where $r_i = f_i/n$. The probability of the event E_i is defined as the limit of the relative frequency:

$$P\left(E_i\right) = \lim_{n \to \infty} \frac{f_i}{n} = \lim_{n \to \infty} r_i = p_i.$$

3. Axiomatic approach

Let $(S, \mathcal{F})$ be a measurable space, where $S = U$ is the sample space and $\mathcal{F}$ is a field of events of S. For $A \in \mathcal{F}$:

1. $0 \leq P(A) \leq 1$;
2. $P(S) = 1$;
3. if $A = \bigcup_i A_i$ with the A_i pairwise disjoint, then

$$P(A) = P\left(\bigcup_i A_i\right) = \sum_i P\left(A_i\right).$$

The triple $(S, \mathcal{F}, P)$ is then a probability space.

Random variables

A random variable is a function which takes its values according to the outcomes of an experiment; for example, in two throws of a coin, $X = 0, 1, 2$ if we are interested in the number of tails.

Distribution functions

Let X be a random variable on the probability space $(S, \mathcal{F}, P)$. For $x \in \mathbb{R}$ define

1. $F_X(x) = P[X \leq x]$;
2. for $a \leq b,\ P[a < X \leq b] = F_X(b) - F_X(a)$.

Then $F_X(x)$ is called the distribution function of X.

Example. Let $A \subset S$, S a sample space, and let $P(A) = p$. Let $X = I_A$, the indicator function of A. Then the distribution function of X is

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1 - p, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$

> *[The notes print the middle case as "$= 1 - p$ if $x \leq x < 1$", i.e. $0 \leq x < 1$.]*

Theorem. The distribution function F_X of a random variable X is non-decreasing and continuous on the right, with

$$F_X(-\infty) = 0 \qquad \text{and} \qquad F_X(\infty) = 1.$$

Proof. Let $x \leq x'$. Then

$$\left[X \leq x'\right] = [X \leq x] + \left[x < X \leq x'\right],$$

the two sets on the right being disjoint, so that

$$P\left[X \leq x'\right] = P[X \leq x] + P\left[x < X \leq x'\right].$$

Also $P\left[x < X \leq x'\right] \geq 0$, hence

$$F_X(x') - F_X(x) \geq 0.$$

This shows that F_X is monotone non-decreasing in x.

Let $\{x_n'\}$ be a monotone decreasing sequence with limit x. As $x_n' \to x$ the events $\left(x < X \leq x_n'\right)$ decrease to $\varnothing$, so

$$F_X\left(x_n'\right) - F_X(x) \to 0.$$

This holds for every such sequence, hence F_X is continuous on the right.

Define

$$A_n = [X \leq n], \qquad n = 1, 2, \ldots$$

Then $A_n \to U$ as $n \to \infty$, so

$$P\left(A_n\right) \to P(U) = 1 \quad \text{as } n \to \infty, \qquad \text{that is } F_X(\infty) = 1.$$

Define

$$B_n = [X \leq -n], \qquad n = 1, 2, \ldots$$

Then $B_n \downarrow \varnothing$ as $n \to \infty$, so

$$P\left(B_n\right) \to 0 \quad \text{as } n \to \infty, \qquad \text{that is } F_X(-\infty) = 0. \qquad \blacksquare$$

1.4.1 EXPECTATION

Let X be a random variable defined on a probability space $(S, \mathcal{F}, P)$ and let $\{A_i;\ i = 1, 2, \ldots, n\}$ be a partition of S with

$$X = \sum_{i=1}^{n} x_i I_{A_i}.$$

Then

$$E(X) = E\left(\sum_{i=1}^{n} x_i I_{A_i}\right) = \sum_{i=1}^{n} x_i E\left(I_{A_i}\right) = \sum_{i=1}^{n} x_i P\left(A_i\right) = \sum_{i=1}^{n} x_i p_i, \qquad (1.3)$$

or

$$E(X) = \sum_{i=1}^{n} x_i \, P\left(X = x_i\right).$$

Properties

Linearity. $E(X \pm Y) = E(X) \pm E(Y)$.

Proof. Let

$$X = \sum_{i=1}^{m} x_i I_{A_i} \qquad \text{and} \qquad Y = \sum_{j=1}^{n} y_j I_{B_j},$$

where $\{A_i;\ i = 1, ..., m\}$ and $\{B_j;\ j = 1, ..., n\}$ are partitions of S. Then

$$X \pm Y = \sum_{i=1}^{m} x_i I_{A_i} \pm \sum_{j=1}^{n} y_j I_{B_j} = \sum_{i=1}^{m} \sum_{j=1}^{n} \left(x_i \pm y_j\right) I_{A_i \cap B_j},$$

so that

$$\begin{aligned} E(X \pm Y) &= \sum_{i=1}^{m} \sum_{j=1}^{n} \left(x_i \pm y_j\right) P\left(A_i \cap B_j\right) \\ &= \sum_{i=1}^{m} \sum_{j=1}^{n} x_i P\left(A_i \cap B_j\right) \pm \sum_{i=1}^{m} \sum_{j=1}^{n} y_j P\left(A_i \cap B_j\right) \\ &= \sum_{i=1}^{m} x_i \sum_{j=1}^{n} P\left(A_i \cap B_j\right) \pm \sum_{j=1}^{n} y_j \sum_{i=1}^{m} P\left(A_i \cap B_j\right) \\ &= \sum_{i=1}^{m} x_i P\left(A_i\right) \pm \sum_{j=1}^{n} y_j P\left(B_j\right) = E(X) \pm E(Y). \qquad \blacksquare \end{aligned}$$

Scale preserving. $E(cX) = c\,E(X)$ for any constant c.

Non-negativity. If $X \geq 0$ then $E(X) \geq 0$.

1.5 MOMENTS

Let X be a random variable. The rth moment of X about zero is $E\left(X^r\right)$; the rth absolute moment about zero is $E|X|^r$; and the rth absolute moment of X about a point a is $E|X - a|^r$.

Lemma. If $E|X|^r < \infty$ and $E|X|^s < \infty$ with $0 < s < r$, then

$$E|X|^s \leq 1 + E|X|^r.$$

Proof. For $0 < s < r$,

$$|X|^s \leq 1 \ \text{ if } |X| \leq 1, \qquad \text{and} \qquad |X|^s \leq |X|^r \ \text{ if } |X| \geq 1.$$

Hence, for every outcome,

$$|X|^s \leq 1 + |X|^r \qquad (0 < s < r).$$

Taking expectations gives

$$E|X|^s \leq 1 + E|X|^r < \infty. \qquad \blacksquare$$

Lemma (C_r-inequality).

$$E|X + Y|^r \leq C_r E|X|^r + C_r E|Y|^r, \qquad C_r = \begin{cases} 1, & r \leq 1, \\ 2^{r-1}, & r \geq 1. \end{cases}$$

Proof (case $r \leq 1$). Let $a > 0$ and $b > 0$. Then

$$\left(\frac{a}{a+b}\right)^r + \left(\frac{b}{a+b}\right)^r \geq 1 \qquad \text{for } r \leq 1,$$

since each term is at least the corresponding term of the sum for $r = 1$ when the arguments lie in $(0, 1)$. Multiplying by $(a + b)^r$,

$$(a + b)^r \le a^r + b^r.$$

Taking $a = |X|$, $b = |Y|$ and then expectations,

$$E|X + Y|^r \le E|X|^r + E|Y|^r. \qquad \blacksquare$$

Proof (case $r \ge 1$). Consider the function

$$Q(p) = p^r + (1 - p)^r, \qquad r \ge 1.$$

This function has its minimum at $p = \frac{1}{2}$, where

$$Q\left(\tfrac{1}{2}\right) = \left(\tfrac{1}{2}\right)^r + \left(\tfrac{1}{2}\right)^r = 2^{1-r}.$$

Hence $Q(p) \ge 2^{1-r}$ for all p, that is

$$p^r + (1 - p)^r \ge 2^{1-r}.$$

Putting $p = \frac{a}{a+b}$ and multiplying by $(a + b)^r$,

$$a^r + b^r \ge 2^{1-r}(a + b)^r, \qquad \text{i.e.} \qquad (a + b)^r \le 2^{r-1}a^r + 2^{r-1}b^r.$$

With $a = |X|$ and $b = |Y|$ this gives

$$|X + Y|^r \le 2^{r-1}|X|^r + 2^{r-1}|Y|^r,$$

and taking expectations,

$$E|X + Y|^r \le 2^{r-1}E|X|^r + 2^{r-1}E|Y|^r. \qquad \blacksquare$$

Lemma (Hölder's inequality). If $r > 1$ and $\frac{1}{r} + \frac{1}{s} = 1$, then

$$E|XY| \le \left(E|X|^r\right)^{1/r}\left(E|Y|^s\right)^{1/s}.$$

Proof. Let

$$Q(p) = \frac{p^r}{r} + \frac{p^{-s}}{s}, \qquad 0 < p < \infty.$$

Differentiating, $Q'(p) = p^{r-1} - p^{-s-1} = p^{-s-1}\left(p^{r+s} - 1\right)$, and since $\frac{1}{r} + \frac{1}{s} = 1$ gives $r + s = rs$, the derivative vanishes only at $p = 1$; there Q has its minimum, so $Q(p) \ge Q(1) = \frac{1}{r} + \frac{1}{s} = 1$. Thus for all $p > 0$

$$\frac{p^r}{r} + \frac{p^{-s}}{s} \ge 1.$$

Let $a > 0$, $b > 0$ and put $p = a^{1/s}b^{-1/r}$. Then $p^r = a^{r/s}b^{-1}$ and $p^{-s} = a^{-1}b^{s/r}$, so

$$\frac{a^{r/s}}{r\,b} + \frac{b^{s/r}}{s\,a} \ge 1.$$

Multiplying by ab and using $r + s = rs$, i.e. $\frac{r}{s} = r - 1$ and $\frac{s}{r} = s - 1$,

$$ab \le \frac{a^r}{r} + \frac{b^s}{s},$$

which is Young's inequality. Now put

$$a = \frac{|X|}{\left(E|X|^r\right)^{1/r}}, \qquad b = \frac{|Y|}{\left(E|Y|^s\right)^{1/s}}.$$

Then

$$\frac{|XY|}{\left(E|X|^r\right)^{1/r}\left(E|Y|^s\right)^{1/s}} \le \frac{1}{r}\cdot\frac{|X|^r}{E|X|^r} + \frac{1}{s}\cdot\frac{|Y|^s}{E|Y|^s}.$$

Taking expectations and using $E|X|^r/E|X|^r = E|Y|^s/E|Y|^s = 1$,

$$\frac{E|XY|}{\left(E|X|^r\right)^{1/r}\left(E|Y|^s\right)^{1/s}} \le \frac{1}{r} + \frac{1}{s} = 1,$$

which is the assertion. $\blacksquare$

Lemma (Minkowski's inequality). For $r \ge 1$,

$$\left(E|X + Y|^r\right)^{1/r} \le \left(E|X|^r\right)^{1/r} + \left(E|Y|^r\right)^{1/r}.$$

Proof. Write

$$E|X + Y|^r = E\left(|X + Y||X + Y|^{r-1}\right) \le E|X||X + Y|^{r-1} + E|Y||X + Y|^{r-1}.$$

Apply Hölder's inequality to each term with exponents r and $s = \frac{r}{r-1}$:

$$E|X||X+Y|^{r-1} \le \left(E|X|^r\right)^{1/r}\left(E|X+Y|^{(r-1)s}\right)^{1/s},$$

and the same with Y in place of X. Since $(r-1)s = r$ and $1/s = 1 - \frac{1}{r}$,

$$E|X+Y|^r \le \left[\left(E|X|^r\right)^{1/r} + \left(E|Y|^r\right)^{1/r}\right]\left(E|X+Y|^r\right)^{1-1/r}.$$

Dividing by $\left(E|X+Y|^r\right)^{1-1/r}$ gives the result. ■

Convex functions

Let f be a real-valued Borel function defined on an open interval I of $\mathbb{R}$, and let (x, x') be a pair of points of I. Then f is a **convex function** if

$$f\left(\frac{x+x'}{2}\right) \le \frac{1}{2}f(x) + \frac{1}{2}f\left(x'\right).$$

Equivalently, for each $x_0 \in I$ there is a number $\lambda(x_0)$ such that

$$\lambda(x_0) \le \frac{f(x) - f(x_0)}{x - x_0}, \qquad x \ne x_0,$$

that is, the graph of f lies above the line through $(x_0, f(x_0))$ with slope $\lambda(x_0)$:

$$f(x) \ge f(x_0) + \lambda(x_0)(x - x_0).$$

Theorem (Jensen's inequality).

$$f(E(X)) \le E(f(X)).$$

Proof. In the inequality above replace x by X and x_0 by $E(X)$:

$$f(X) \ge f(E(X)) + \lambda(E(X))(X - E(X)).$$

Taking expectations,

$$E(f(X)) \ge f(E(X)) + \lambda(E(X))(E(X) - E(X)) = f(E(X)). \qquad ■$$

The basic inequality

Theorem. Let X be an arbitrary random variable and let g be an even, non-negative, non-decreasing Borel function on the half-open interval $[0, \infty)$. Then for every $a > 0$,

$$\frac{E[g(X)] - g(a)}{\sup g(x)} \le P[g(X) > a] \le \frac{E[g(X)]}{g(a)}. \qquad (1.4)$$

> *[The notes print this as $\frac{Eg(x) - g(a)}{\sup g(x)} \le p[(x) > a] \le \frac{Eg(x)}{g(a)}$; the random variable is X and the event is $[g(X) > a]$.]*

Proof. $g(X)$ is a measurable function, since g is a Borel function on $\mathbb{R}$. Being non-negative, its integral exists, and for $A = [g(X) > a]$

$$E[g(X)] = \int_A g(x)\,dP + \int_{A^c} g(x)\,dP.$$

Inside A, g is non-decreasing and $g(a) \le g(x) \le \sup g(x)$, so

$$g(a)\,P(A) \le \int_A g(x)\,dP \le \sup g(x)\,P(A). \qquad (1.5)$$

Inside A^c, $0 \le g(x) \le g(a)$, hence

$$0 \le \int_{A^c} g(x)\,dP \le g(a)\,P(A^c) \le g(a). \qquad (1.6)$$

Adding (1.5) and (1.6),

$$g(a)\,P(A) \le \int_A g(x)\,dP + \int_{A^c} g(x)\,dP \le \sup g(x)\,P(A) + g(a), \qquad (1.7)$$

that is

$$g(a)\,P(A) \le E[g(X)] \le \sup g(x)\,P(A) + g(a). \qquad (1.8)$$

From the left-hand inequality, $P(A) \le \frac{E[g(X)]}{g(a)}$; from the right-hand one,

$$\frac{E\big[g(X)\big] - g(a)}{\sup g(x)} \le P(A).$$

Together these give (1.4). ■

Markov's inequality. Put $g(x) = |x|^r$ in the basic inequality. Then, for $a > 0$,

$$P\left(|X| > a\right) \le \frac{E|X|^r}{a^r}.$$

Chebyshev's inequality. In Markov's inequality replace X by $X - \mu$ and put $r = 2$. Then, for $\epsilon > 0$,

$$P\big[\big|X - \mu\big| > \epsilon\big] \le \frac{E\left(X - \mu\right)^2}{\epsilon^2} = \frac{\sigma^2}{\epsilon^2},$$

or equivalently

$$P\big[\big|X - \mu\big| \le \epsilon\big] \ge 1 - \frac{\sigma^2}{\epsilon^2}.$$

1.5.1 Convergence in probability

Let $\{X_n\}$ be a sequence of random variables. The sequence is said to converge in probability to the random variable X if

$$\lim_{n \to \infty} P\big[\big|X_n - X\big| > \epsilon\big] = 0 \qquad \text{for every } \epsilon > 0,$$

or equivalently

$$\lim_{n \to \infty} P\big[\big|X_n - X\big| \le \epsilon\big] = 1.$$

This is written $X_n \xrightarrow{p} X$.

Theorem. $X_n \xrightarrow{p} 0$ if $E\big|X_n\big|^r \to 0$. The theorem therefore provides a sufficient condition for convergence in probability.

Proof. In Markov's inequality put $X = X_n$. Then, for $\epsilon > 0$,

$$P\big[\big|X_n\big| > \epsilon\big] \le \frac{E\big|X_n\big|^r}{\epsilon^r}. \qquad (1.9)$$

Obviously, if $E\big|X_n\big|^r \to 0$ as $n \to \infty$, then $P\big[\big|X_n\big| > \epsilon\big] \to 0$ as $n \to \infty$. ■

Theorem.

$$X_n \xrightarrow{p} 0 \quad \Leftrightarrow \quad E\left(\frac{|X_n|}{1 + |X_n|}\right) \to 0 \quad \text{as } n \to \infty.$$

Proof. For any random variable X, the random variable $\frac{|X|}{1 + |X|}$ is bounded by unity; that is,

$$g(x) = \frac{|X|}{1 + |X|}, \qquad \sup g(x) = 1.$$

Put $g\left(X_n\right) = \frac{|X_n|}{1 + |X_n|}$ in the basic inequality (1.4), taking $a = \epsilon$:

$$E\big[g\left(X_n\right)\big] - \epsilon \le P\big[g\left(X_n\right) > \epsilon\big] \le \frac{E\big[g\left(X_n\right)\big]}{\epsilon}. \qquad (1.10)$$

From the right-hand side: if $E\big[g\left(X_n\right)\big] \to 0$ as $n \to \infty$, then

$$P\left[|X_n| > \frac{\epsilon}{1 - \epsilon}\right] = P\big[g\left(X_n\right) > \epsilon\big] \to 0 \quad \text{as } n \to \infty.$$

From the left-hand side: if $P\big[\big|X_n\big| > \epsilon\big] \to 0$ as $n \to \infty$, then

$$\left(E\left[\frac{|X|}{1 + |X|}\right] - \epsilon\right) \le P\big[g\left(X_n\right) > \epsilon\big] \to 0 \quad \text{as } n \to \infty.$$

But $\frac{|X|}{1 + |X|}$ is positive and ϵ is arbitrary, so in the limit

$$E\left[\frac{|X|}{1 + |X|}\right] \to 0 \qquad \text{as } n \to \infty. \qquad ■$$

1.5.2 Convergence almost surely

A sequence $\{X_n\}$ of random variables is said to converge to X **almost surely** if $X_n(w) \to X(w)$ for all w except the points of a null set v with $P(v) = 0$. This is written $X_n \to X$ $(a.s.)$. Convergence almost surely is also called **strong convergence**.

Corollary. Convergence almost surely implies convergence in probability.

1.5.3 Convergence in distribution

Let X_n be a random variable with distribution function $F_n(x)$, and let X be a random variable with distribution function $F(x)$. The sequence $\{X_n\}$ is said to converge in distribution to X if

$$F_n(x) \to F(x) \qquad \text{as } n \to \infty$$

at every point x at which F is continuous. This convergence is also called convergence in law or **weak convergence**.

Theorem. Convergence in probability implies convergence in distribution.

Proof. Let $x' < x < x''$. Write

$$\left[X \le x'\right] = \left[X_n \le x,\ X \le x'\right] + \left[X_n > x,\ X \le x'\right] \subset \left[X_n \le x\right] + \left[X_n > x,\ X \le x'\right].$$

Therefore

$$P\left[X \le x'\right] \le P\left[X_n \le x\right] + P\left[X_n > x,\ X \le x'\right],$$

or

$$F\left(x'\right) \le F_n(x) + P\left[X_n > x,\ X \le x'\right].$$

But

$$P\left[X_n > x,\ X \le x'\right] \le P\left[\left|X_n - X\right| > x - x'\right] \to 0 \qquad \text{as } n \to \infty,$$

because of convergence in probability. Hence

$$F\left(x'\right) \le \liminf_{n \to \infty} F_n(x). \qquad (1.11)$$

On the other hand,

$$\left[X_n \le x\right] = \left[X \le x'',\ X_n \le x\right] + \left[X > x'',\ X_n \le x\right] \subset \left[X \le x''\right] + \left[X > x'',\ X_n \le x\right],$$

so that

$$P\left[X_n \le x\right] \le P\left[X \le x''\right] + P\left[X > x'',\ X_n \le x\right],$$

or

$$F_n(x) \le F\left(x''\right) + P\left[\left|X_n - X\right| > x'' - x\right] \to F\left(x''\right) \qquad \text{as } n \to \infty.$$

Hence

$$\limsup_{n \to \infty} F_n(x) \le F\left(x''\right). \qquad (1.12)$$

Letting $x' \uparrow x$ and $x'' \downarrow x$ through points of continuity of F, (1.11) and (1.12) give

$$F_n(x) \to F(x) \qquad \text{as } n \to \infty \quad \text{whenever } X_n \xrightarrow{p} X. \qquad (1.13) \qquad \blacksquare$$

1.5.4 Convergence in the rth mean

A sequence of random variables X_n is said to converge in the rth mean to X if

$$E\left|X_n - X\right|^r \to 0 \qquad \text{as } n \to \infty,$$

written $X_n \xrightarrow{r} X$.

Theorem. If X_n converges in the rth mean to X, then $X_n \xrightarrow{p} X$, and moreover $E\left|X_n\right|^r \to E|X|^r$.

Proof (case $r \le 1$). In the C_r-inequality

$$E|X + Y|^r \le E|X|^r + E|Y|^r$$

replace X by $X_n - X$ and Y by X. Then

$$E\left|X_n\right|^r \le E\left|X_n - X\right|^r + E|X|^r. \qquad (1.14)$$

If $X_n \xrightarrow{r} X$ then $E|X_n - X|^r \to 0$ as $n \to \infty$, and from (1.14)

$$E|X_n|^r - E|X|^r \to 0 \quad \text{as } n \to \infty, \qquad \text{that is} \quad E|X_n|^r \to E|X|^r.$$

Proof (case $r \geq 1$). In Minkowski's inequality

$$\left(E|X+Y|^r\right)^{1/r} \leq \left(E|X|^r\right)^{1/r} + \left(E|Y|^r\right)^{1/r}$$

replace X by $X_n - X$ and Y by X. Then

$$\left(E|X_n|^r\right)^{1/r} \leq \left(E|X_n - X|^r\right)^{1/r} + \left(E|X|^r\right)^{1/r}.$$

But

$$\left(E|X_n - X|^r\right)^{1/r} = \left[E|X_n - X|^r\right]^{1/r} \to 0 \qquad \text{as } n \to \infty,$$

so

$$\left(E|X_n|^r\right)^{1/r} \to \left(E|X|^r\right)^{1/r},$$

which is the assertion. Convergence in probability then follows from the Markov inequality (1.9). ■

1.5.5 Monotone convergence theorem

Let X_n be positive and let the sequence $\{X_n\}$ be such that it monotonically increases to X. Then

$$E(X_n) \uparrow E(X), \qquad \text{i.e.} \quad X_n \uparrow X \Rightarrow E(X_n) \uparrow E(X).$$

1.6 FATOU'S THEOREM

1. If $Y \leq X_n$ with Y integrable, then

$$E\left(\liminf_n X_n\right) \leq \liminf_n E(X_n).$$

Proof. Let

$$Y_n = \inf_{k \geq n} X_k.$$

Then

$$\lim_n Y_n = \liminf_n X_n \qquad \text{and} \qquad X_n \geq Y_n.$$

Suppose first that $Y_n \geq 0$. The sequence Y_n is increasing with limit $\liminf X_n$; since $0 \leq Y_n \uparrow \liminf X_n$, the monotone convergence theorem gives

$$E(Y_n) \uparrow E\left(\liminf X_n\right).$$

But also $X_n \geq Y_n$, so $E(X_n) \geq E(Y_n)$; taking limits on both sides,

$$\lim_n E(X_n) \geq \lim_n E(Y_n) = E\left(\liminf_n X_n\right).$$

For a general integrable Y, apply the preceding argument to $X_n - Y \geq 0$ and add $E(Y)$ to both sides. ■

2. If $X_n \leq Z$ with Z integrable, then

$$\limsup_n E(X_n) \leq E\left(\limsup_n X_n\right).$$

Proof. Let $X_n \leq Z$ and put $Z_n = Z - X_n$. Then $Z_n \geq 0$, so by part 1

$$\liminf_n E(Z_n) \geq E\left(\liminf_n Z_n\right).$$

But

$$E(Z_n) = E(Z) - E(X_n), \qquad \liminf_n Z_n = Z - \limsup_n X_n,$$

so

$$E(Z) - \limsup_n E(X_n) \geq E(Z) - E\left(\limsup_n X_n\right),$$

which gives the result. ■

3. If $Y \leq X_n \leq Z$ with both Y and Z integrable, and $X_n \to X$ almost surely, then

$$\lim_n E(X_n) = E(X).$$

Proof. Since $X_n \to X$ almost surely,

$$\liminf_n X_n = \limsup_n X_n = X.$$

From parts 1 and 2,

$$E(X) = E\left(\liminf_n X_n\right) \le \liminf_n E\left(X_n\right) \le \limsup_n E\left(X_n\right) \le E\left(\limsup_n X_n\right) = E(X).$$

Hence $\lim_n E\left(X_n\right)$ exists and equals $E(X)$. ■

1.7 DOMINATED CONVERGENCE THEOREM (LEBESGUE)

Let $\{X_n\}$ be a sequence of random variables and suppose that for all n

$$\left|X_n\right| \le Y, \qquad Y \text{ integrable.}$$

Then $\{X_n\}$ converges in probability to X implies

$$E\left(X_n\right) \to E(X).$$

1.7.1 Product space

Cartesian product. Let A and B be sets. The set $A \times B$ consisting of all pairs $\left(w_1, w_2\right)$ with $w_1 \in A$ and $w_2 \in B$ is called the Cartesian product of A and B. If $A = \mathbb{R}$ and $B = \mathbb{R}$ then

$$A \times B = \mathbb{R} \times \mathbb{R} = \mathbb{R}^2,$$

the Cartesian plane.

Consider the measurable spaces $\left(U_1, \mathcal{F}_1\right)$ and $\left(U_2, \mathcal{F}_2\right)$. Then

$$\left(U_1 \times U_2,\ \mathcal{F}_1 \times \mathcal{F}_2\right)$$

is the **product measurable space**. Also suppose that $\left(U_1, \mathcal{F}_1, \mu_1\right)$ and $\left(U_2, \mathcal{F}_2, \mu_2\right)$ are measure spaces. Let $A_1 \in \mathcal{F}_1$ and $A_2 \in \mathcal{F}_2$ and consider the measure λ such that

$$\lambda\left(A_1 \times A_2\right) = \mu_1\left(A_1\right)\mu_2\left(A_2\right).$$

Then

$$\left(U_1 \times U_2,\ \mathcal{F}_1 \times \mathcal{F}_2,\ \lambda\right)$$

is the product measure space.

Let $A \subset U_1 \times U_2$. Then

$$A_{w_1} = \left\{w_2 : \left(w_1, w_2\right) \in A\right\}$$

is called the w_1-section of A, and similarly

$$A_{w_2} = \left\{w_1 : \left(w_1, w_2\right) \in A\right\}$$

is called the w_2-section of A.

Let f be a function defined on $U_1 \times U_2$. For $w_1 \in U_1$, the w_1-section of f is the function f_{w_1} defined on A_{w_1} by

$$f_{w_1}\left(w_2\right) = f\left(w_1, w_2\right),$$

and similarly

$$f_{w_2}\left(w_1\right) = f\left(w_1, w_2\right).$$

1.7.2 Mutual convergence theorem

If $X_n \xrightarrow{p} X$, then $X_n - X_m \xrightarrow{p} 0$ as $n, m \to \infty$.

Proof. Write

$$\left[\left|X_n - X_m\right| > \epsilon\right] \subset \left[\left|X_n - X\right| > \frac{\epsilon}{2}\right] \cup \left[\left|X_m - X\right| > \frac{\epsilon}{2}\right]. \qquad (1.15)$$

This implies

$$P\left[\left|X_n - X_m\right| > \epsilon\right] \le P\left[\left|X_n - X\right| > \frac{\epsilon}{2}\right] + P\left[\left|X_m - X\right| > \frac{\epsilon}{2}\right].$$

Both terms on the right converge to zero because of convergence in probability, hence the proof. ■

1.7.3 Closure under arithmetic operations

Theorem. Let $X_n \xrightarrow{p} X$, $Y_n \xrightarrow{p} Y$ and $a \in \mathbb{R}$. Then

1. $aX_n \xrightarrow{p} aX$ (scalar multiple);
2. $X_n + Y_n \xrightarrow{p} X + Y$ (additive);
3. $X_n Y_n \xrightarrow{p} XY$ (multiplicative);
4. $\frac{X_n}{Y_n} \xrightarrow{p} \frac{X}{Y}$ (quotient), where $P\left(Y_n = 0\right) = 0$ and $P(Y = 0) = 0$.

Proof.

1. If $a = 0$ the result is trivially true. Suppose $a \neq 0$; then for $\epsilon > 0$

$$P\left[\left|aX_n\right| > \epsilon\right] = P\left[|a|\left|X_n\right| > \epsilon\right] = P\left[\left|X_n\right| > \frac{\epsilon}{|a|}\right] \to 0 \qquad \text{as } n \to \infty. \qquad \blacksquare$$

2. Write

$$\begin{aligned} P\left[\left|X_n + Y_n - X - Y\right| > \epsilon\right] &\leq P\left[\left|X_n - X\right| + \left|Y_n - Y\right| > \epsilon\right] \\ &\leq P\left[\left|X_n - X\right| > \frac{\epsilon}{2}\right] + P\left[\left|Y_n - Y\right| > \frac{\epsilon}{2}\right]. \end{aligned}$$

The terms on the right tend to 0 as $n \to \infty$, hence the proof. ■

3. Write

$$\begin{aligned} P\left[\left|X_n Y_n - XY\right| > \epsilon\right] &= P\left[\left|X_n Y_n - X_n Y + X_n Y - XY\right| > \epsilon\right] \\ &= P\left[\left|X_n\left(Y_n - Y\right) + Y\left(X_n - X\right)\right| > \epsilon\right] \\ \leq P\left[\left|X_n\left(Y_n - Y\right)\right| > \frac{\epsilon}{2}\right] &+ P\left[\left|Y\left(X_n - X\right)\right| > \frac{\epsilon}{2}\right]. \qquad (1.16) \end{aligned}$$

For the first term, let $\sigma > 0$. Then

$$\begin{aligned} \left[\left|X_n\left(Y_n - Y\right)\right| > \frac{\epsilon}{2}\right] &= \left[\left|X_n\left(Y_n - Y\right)\right| > \frac{\epsilon}{2},\ \left|Y_n - Y\right| > \sigma\right] \cup \left[\left|X_n\left(Y_n - Y\right)\right| > \frac{\epsilon}{2},\ \left|Y_n - Y\right| \leq \sigma\right] \\ &\subset \left[\left|Y_n - Y\right| > \sigma\right] \cup \left[\left|X_n\right| > \frac{\epsilon}{2\sigma}\right], \end{aligned}$$

and, writing $\eta = \frac{\epsilon}{2\sigma} > 0$,

$$\left[\left|X_n\right| > \eta\right] \subset \left[\left|X_n - X\right| > \eta\right] \cup \left[|X| > \eta\right],$$

so

$$P\left[\left|X_n\left(Y_n - Y\right)\right| > \frac{\epsilon}{2}\right] \leq P\left[\left|X_n - X\right| > \eta\right] + P\left[\left|Y_n - Y\right| > \sigma\right] + P\left[|X| > \eta\right].$$

As $n \to \infty$ the first two terms tend to 0; the last one can be made as small as we please by taking η large, because X is finite almost surely. For the second term in (1.16),

$$P\left[\left|Y\left(X_n - X\right)\right| > \frac{\epsilon}{2}\right] = P\left[|Y|\left|X_n - X\right| > \frac{\epsilon}{2}\right] \to 0 \qquad \text{as } n \to \infty,$$

since $|Y| < \infty$ almost surely and $X_n \to X$ in probability. Hence both terms on the right of (1.16) tend to 0 as $n \to \infty$, which completes the proof. ■

4. The proof is similar to that of 3, provided $Y_n \neq 0$ and $Y \neq 0$; write

$$\frac{X_n}{Y_n} - \frac{X}{Y} = \frac{X_n Y - XY_n}{Y_n Y} = \frac{\left(X_n - X\right)Y + X\left(Y - Y_n\right)}{Y_n Y}$$

and argue as in part 3, using $P\left(Y_n = 0\right) = P(Y = 0) = 0$. ■

1.8 FUBINI'S THEOREM

1. A necessary and sufficient condition that a measurable subset A of $U_1 \times U_2$ have measure 0 is that μ_1-almost every w_1-section (or μ_2-almost every w_2-section) of A have measure zero.

2. If h is a non-negative measurable function on $U_1 \times U_2$, then

$$\int h\,d\lambda = \iint h\,d\mu_1\,d\mu_2 = \iint h\,d\mu_2\,d\mu_1.$$

3. If h is integrable on $U_1 \times U_2$, then almost every section of h is integrable. If

$$f(w_1) = \int h\,d\mu_2(w_2), \qquad g(w_2) = \int h\,d\mu_1(w_1),$$

then f and g are integrable and

$$\int h\,d\lambda = \int f\,d\mu_1 = \int g\,d\mu_2.$$

1.9 CHARACTERISTIC FUNCTIONS

Let X be a random variable with distribution function $F(x)$. Its **characteristic function** is given by

$$Q_X(t) = E\left(e^{itX}\right) = \int e^{itx}\,dF(x).$$

The characteristic function has two parts, real and imaginary, because

$$e^{itx} = \cos(tx) + i\sin(tx).$$

This implies

$$Q_X(t) = \int \cos(tx)\,dF(x) + i\int \sin(tx)\,dF(x).$$

1.9.1 Properties

1. The characteristic function $Q_X(t)$ of a random variable X is continuous.

Proof. Let t and t' be parameters. Then

$$\left|Q(t') - Q(t)\right| = \left|\int \left(e^{it'x} - e^{itx}\right)dF(x)\right| \le \int \left|e^{it'x} - e^{itx}\right|dF(x).$$

But

$$\left|e^{it'x} - e^{itx}\right| = \left|e^{it'x}\right|\left|1 - e^{i(t-t')x}\right| \to 0 \qquad \text{as } t - t' \to 0,$$

and also

$$\left|e^{it'x} - e^{itx}\right| \le 2.$$

By the dominated convergence theorem,

$$\left|Q_X(t') - Q_X(t)\right| \to 0 \qquad \text{as } \left|t - t'\right| \to 0,$$

which implies that $Q_X(t)$ is continuous. ■

2. $\left|Q(t)\right| \le 1$ **(bounded).**

This is mainly because

$$\left|e^{itx}\right|^2 = \cos^2(tx) + \sin^2(tx) = 1.$$

3. If $Q_X(t)$ is the characteristic function of X, then for $Y = a + bX$

$$Q_Y(t) = Q_X(bt)\,e^{iat} \qquad \text{(linear combination).}$$

Lemma. For any characteristic function,

$$\mathrm{Re}\left(1 - Q(t)\right) \ge \frac{1}{4}\mathrm{Re}\left(1 - Q(2t)\right),$$

where $\mathrm{Re}Q(t)$ is the real part of $Q(t)$.

Proof. Using the reality of the distribution function,

$$1 - \mathrm{Re}Q(2t) = \int \left(1 - \cos(2tx)\right)dF(x) = 2\int \left(1 - \cos^2(tx)\right)dF(x)$$
$$= 2\int \left(1 - \cos(tx)\right)\left(1 + \cos(tx)\right)dF(x).$$

But $1 + \cos(tx) \le 2$, so

$$1 - \mathrm{Re}Q(2t) \le 4\int \big(1 - \cos(tx)\big)\, dF(x) = 4\mathrm{Re}\big(1 - Q(t)\big),$$

that is

$$\mathrm{Re}\big(1 - Q(t)\big) \ge \frac{1}{4}\mathrm{Re}\big(1 - Q(2t)\big). \qquad \blacksquare$$

1.9.2 Inversion theorem

Let $Q(t)$ be the characteristic function of X, whose distribution function is F, and let a, b be points of continuity of F with $a < b$. Then

$$P\,(\,a < X \le b\,) = F(b) - F(a) = \lim_{T\to\infty}\frac{1}{2\pi}\int_{-T}^{T}\frac{e^{-ita} - e^{-itb}}{it}\,Q(t)\,dt.$$

1.9.3 Independence

Let A and B be two events with probabilities $P(A)$ and $P(B)$. The compound event AB has probability $P(AB)$ satisfying

$$P(AB) = P(A)\,P\,(\,B \mid A\,).$$

$P\,(\,B \mid A\,)$ is called the conditional probability of B occurring given that A has occurred:

$$P\,(\,B \mid A\,) = \frac{P(AB)}{P(A)}.$$

If B and A are independent then the probability becomes unconditional, that is $P\,(\,B \mid A\,) = P(B)$, and

$$P(AB) = P(A)\,P(B).$$

Note that if A and B are independent then A and B^c are also independent. Mutually exclusive events, on the other hand, are always dependent on each other: suppose A and B are mutually exclusive, so that $P(AB) = 0$; if $P(A) \ne 0$ and $P(B) \ne 0$, independence would give $P(AB) = P(A)P(B) \ne 0$, which contradicts $P(AB) = 0$.

The null set is independent of any other set. Let N be the null set with $P(N) = 0$ and let A be any other arbitrary event. Then

$$P(AN) = P(A)\,P\,(\,N \mid A\,) = P(A)\cdot 0 = 0 = P(A)\,P(N),$$

so an impossible (null) event is independent of any other event. Also

$$P(NN) = P(N)\,P\,(\,N \mid N\,) = 0\cdot 0 = P(N)\,P(N),$$

so N is independent of itself.

The sure event is independent of any other set. Let N^c be the sure event with $P\left(N^c\right) = 1$ and let A be any arbitrary event. Then

$$P\left(AN^c\right) = P(A)\,P\left(N^c \mid A\right) = P(A)\cdot 1 = P(A)\,P\left(N^c\right),$$

so N^c is independent of any other event. Also

$$P\left(N^cN^c\right) = P\left(N^c\right)P\left(N^c \mid N^c\right) = 1\cdot 1 = P\left(N^c\right)P\left(N^c\right),$$

so the sure event is independent of itself.

1.9.4 Class of independent events

Let $\mathcal{C} = \{A_1, A_2, \ldots, A_n\}$ be a class of events and let $\left\{A_{j_1}, A_{j_2}, \ldots, A_{j_k}\right\}$ be a sub-class of $\mathcal{C}$. Then $\mathcal{C}$ is a **mutually independent class** if, for every $k \le n$,

$$P\left(A_{j_1}A_{j_2}\cdots A_{j_k}\right) = \prod_i P\left(A_{j_i}\right).$$

1.9.5 Independent classes

Consider two classes $\mathcal{C}_1$ and $\mathcal{C}_2$. The two classes are independent if, for every $A_i \in \mathcal{C}_1$ and $B_j \in \mathcal{C}_2$,

$$P\left(A_iB_j\right) = P\left(A_i\right)P\left(B_j\right).$$

If there are many classes $\mathcal{C}_1, \mathcal{C}_2, \ldots, \mathcal{C}_q$, they are independent if, for $A_{i_1} \in \mathcal{C}_1, A_{i_2} \in \mathcal{C}_2, \ldots, A_{i_q} \in \mathcal{C}_q$,

$$P\left(A_{i_1}A_{i_2}\cdots A_{i_q}\right) = \prod_j P\left(A_{i_j}\right).$$

1.9.6 Independent random variables

Consider the set $\{X_t : t = 1, 2, \ldots, T\}$ of random variables. The set is said to be a mutually independent set of random variables if the σ-fields induced by the random variables are mutually independent.

1.9.7 σ-field induced by a sequence of random variables

Let $X_1, X_2, \ldots, X_n$ be a sequence of random variables. The σ-field induced by $(X_1, X_2, \ldots, X_n)$ is the minimal σ-field containing all the σ-fields induced by each of them:

$$\mathcal{A}_n = \sigma\left[\bigcup_i X_i^{-1}\mathcal{B}\right].$$

1.9.8 Tail σ-field

Let D_n be the σ-field induced by $(X_n, X_{n+1}, \ldots)$. Thus

$$D_n = \sigma\left[\bigcup_{i \geq n} X_i^{-1}\mathcal{B}\right].$$

This σ-field contains D_{n+1}, so D_n is a monotone decreasing sequence of σ-fields whose limit is

$$\tau = \bigcap_{n=1}^{\infty} D_n.$$

τ is a σ-field because it is an intersection of σ-fields. It is usually referred to as a **tail σ-field** and its events are called tail events. Any σ-field D_n contains it.

1.10 KOLMOGOROV'S 0–1 LAW

It states that the tail events of a sequence of independent random variables have probabilities 0 or 1.

Proof. The tail σ-field τ is contained in D_n by definition. Also, the σ-field induced by $(X_1, X_2, \ldots, X_n)$, denoted by $\mathcal{B}(X_1, X_2, \ldots, X_n)$, is independent of D_{n+1}, and τ is contained in D_{n+1}; hence τ is independent of $\mathcal{B}$. But $\mathcal{B}$ is a monotone increasing set of σ-fields whose limit is

$$\bigcup_{n=1}^{\infty} \mathcal{B}(X_1, X_2, \ldots, X_n),$$

that is

$$\mathcal{B}(X_1, X_2, \ldots, X_n) \uparrow \bigcup_{n=1}^{\infty} \mathcal{B}(X_1, X_2, \ldots, X_n) = D_1,$$

which contains τ. This implies that D_1 and τ are independent, and hence that τ is independent of itself. Therefore the events of τ are independent of themselves, so each of them is either a null event or a sure event. ■

1.10.1 Borel–Cantelli lemma

This lemma has two parts.

1. Let $\{A_n\}$ be a sequence of arbitrary events. If $\sum_{n=1}^{\infty} P(A_n) < \infty$, then the probability that the event A_n occurs infinitely often is zero:

$$P\left(\limsup_n A_n\right) = 0.$$

Proof. We know that

$$P\left(\bigcup_{k=r}^{s} A_k\right) \le \sum_{k=r}^{s} P\left(A_k\right)$$

irrespective of whether the A_k are independent or not. Then

$$P\left(\bigcup_{k=r}^{s} A_k\right) \le \sum_{k=r}^{s} P\left(A_k\right) \to 0 \qquad \text{as } r,s \to \infty \text{ if } \sum_{n} P\left(A_n\right) < \infty,$$

which implies that

$$\lim_{r}\lim_{s} P\left(\bigcup_{k=r}^{s} A_k\right) = 0 = P\left(\limsup_{n} A_n\right). \qquad \blacksquare$$

2. Let $\left\{B_n\right\}$ be a sequence of independent events. If $\sum_{n=1}^{\infty} P\left(B_n\right) = \infty$, then the probability that B_n occurs infinitely often is 1:

$$P\left(\limsup_{n} B_n\right) = 1.$$

Proof. Write

$$P\left(\limsup_{n} B_n\right) = \lim_{r}\lim_{s}\left(1 - P\left(\bigcap_{k=r}^{s} B_k^c\right)\right) = \lim_{r}\lim_{s}\left[1 - \prod_{k=r}^{s}\left(1 - P\left(B_k\right)\right)\right],$$

because $\left\{B_k^c\right\}$ is an independent sequence and therefore

$$P\left(\bigcap_{k=r}^{s} B_k^c\right) = \prod_{k=r}^{s}\left(1 - P\left(B_k\right)\right).$$

But for $0 \le P\left(B_k\right) \le 1$,

$$1 - P\left(B_k\right) \le e^{-P\left(B_k\right)},$$

so that

$$\prod_{k=r}^{s}\left(1 - P\left(B_k\right)\right) \le e^{-\sum_{k=r}^{s} P\left(B_k\right)}.$$

If $\sum_k P\left(B_k\right) = \infty$ then $\sum_{k=r}^{s} P\left(B_k\right)$ is infinite in the limit $s \to \infty$ for every r, so

$$\lim_{s} e^{-\sum_{k=r}^{s} P\left(B_k\right)} = 0,$$

and hence

$$1 - \prod_{k=r}^{s}\left(1 - P\left(B_k\right)\right) \to 1, \qquad \text{that is} \quad P\left(\limsup_{n} B_n\right) = 1. \qquad \blacksquare$$

1.10.2 Equivalent sequences

Two sequences $\left\{X_n\right\}$ and $\left\{X_n^{'}\right\}$ are **tail-equivalent** if they differ only on a finite number of terms: that is, for almost all $w \in U$ there exists a finite number $n_0(w)$ such that for all $n \ge n_0(w)$

$$X_n(w) = X_n^{'}(w).$$

Let

$$S_n = \sum_{k=1}^{n} X_k, \qquad S_n^{'} = \sum_{k=1}^{n} X_k^{'}.$$

If $\sum_n P\left(X_n \ne X_n^{'}\right) < \infty$ then $\left\{X_n\right\}$ and $\left\{X_n^{'}\right\}$ are tail-equivalent, and S_n and $S_n^{'}$ are **convergent-equivalent**.

1.10.3 Convergence in probability of S_n

Suppose that in the basic inequality X is replaced by $S_n - E\left(S_n\right)$ and $g(x) = x^2$. Assume that $X_1, X_2, \ldots, X_n$ are mutually independent, so that

$$E\left(S_n\right) = E\left(\sum_{k=1}^{n} X_k\right) = \sum_{k=1}^{n} E\left(X_k\right).$$

Let $V\left(X_k\right) = \sigma_k^2$. Then, for $a = \epsilon$ and $g(a) = \epsilon^2$, the basic inequality gives

$$\frac{E\left[S_n - E\left(S_n\right)\right]^2 - \epsilon^2}{\sup\left[S_n - E\left(S_n\right)\right]^2} \le P\left[\left|S_n - E\left(S_n\right)\right| > \epsilon\right] \le \frac{E\left(S_n - E\left(S_n\right)\right)^2}{\epsilon^2}.$$

Since, by independence,

$$E\left(S_n - E\left(S_n\right)\right)^2 = V\left(S_n\right) = \sum_{k=1}^{n} \sigma_k^2,$$

we obtain

$$P\left[\left|S_n - E\left(S_n\right)\right| > \epsilon\right] \le \frac{1}{\epsilon^2} \sum_{k=1}^{n} \sigma_k^2.$$

If $\sum_k \sigma_k^2 < \infty$ then $S_n - E\left(S_n\right)$ converges in probability (to zero).

1.10.4 Stability of independent random variables

A sequence $\left\{X_n\right\}$ of independent random variables is said to be **stable in probability** if the sequence

$$\frac{S_n - a_n}{b_n}$$

converges in probability to zero, where $\left\{a_n\right\}$ and $\left\{b_n\right\}$ are numerical sequences with $b_n > 0$.

Corollary. If $\frac{1}{b_n^2}\sum_k \sigma_k^2 \to 0$, then

$$\frac{S_n - E\left(S_n\right)}{b_n} \xrightarrow{p} 0.$$

1.11 BERNOULLI'S WEAK LAW OF LARGE NUMBERS

Suppose $\left\{X_k\right\}$ is a sequence of independent Bernoulli random variables with

$$P\left(X_k = 0\right) = 1 - p, \qquad P\left(X_k = 1\right) = p.$$

For $S_n = \sum_{k=1}^{n} X_k$,

$$E\left(S_n\right) = np, \qquad V\left(S_n\right) = np(1-p) = npq.$$

From Chebyshev's inequality, as $n \to \infty$,

$$P\left[\left|\frac{S_n}{n} - p\right| > \epsilon\right] \le \frac{npq}{n^2\epsilon^2} \le \frac{1}{4n\epsilon^2} \to 0.$$

Thus

$$\frac{S_n - np}{n} \xrightarrow{p} 0, \qquad \text{or} \qquad \frac{S_n}{n} \xrightarrow{p} p.$$

This stability property is called the **Bernoulli weak law of large numbers**. If instead $\frac{S_n}{n} \to p$ almost surely, the stability property is the **Borel strong law of large numbers**.

1.11.1 Central limit theorem

Let $X_1, X_2, \ldots, X_n$ be independently distributed random variables and let $\bar{X}$ be the mean of the random variables. Then the distribution of $\bar{X}$ converges to $N\left(\mu, \sigma^2\right)$, that is

$$\bar{X} \sim N\left(\mu, \sigma^2\right) \qquad \text{approximately, for large } n.$$

1.11.2 Theorem (Lindeberg–Lévy)

Let $\left\{X_k\right\}$ be a sequence of independent and identically distributed random variables with

$$E\left(X_k\right) = 0, \qquad V\left(X_k\right) = \sigma^2 < \infty.$$

Then, as $n \to \infty$,

$$\frac{\sqrt{n}\,\bar{X}}{\sigma} \to N(0,1) \qquad \text{in distribution.}$$

1.11.3 Theorem (Liapounov)

Let $\{X_n\}$ be a sequence of random variables with $V(X_k) = \sigma_k^2$ and let

$$g_n^2 = \sum_{k=1}^{n} \sigma_k^2.$$

If

$$\frac{1}{g_n^{2+\theta}} \sum_{k=1}^{n} E\left[|X_k|^{2+\theta}\right] \to 0$$

for some θ with $0 < \theta < 1$ as $n \to \infty$, then

$$\frac{S_n}{g_n} \to N(0,1) \qquad \text{in distribution.}$$

APPENDIX A — CORRECTIONS MADE TO THE SOURCE

Every page of the source is reproduced above. The following points are places where the notes as printed are incomplete or mistyped; each has been corrected in the text and is listed here so that the correction can be checked against the original.

#	Source page	As printed	As corrected
1	3	limit of $A_n = \left\{w: 2 - \frac{1}{n} < w < 7 - \frac{2}{n}\right\}$ given as $\{w: 2 < w \le 7\}$	$\{w: 2 \le w < 7\}$ (endpoint inequalities the other way round)
2	3	$\{w: 2 < w\,6\}$	$\{w: 2 < w < 6\}$
3	4	$A^c \in A$	$A^c \in \mathcal{F}$
4	4	$\{A^c \in B^c\}^c$	$(A^c \cap B^c)^c$
5	8	$p(0) = 1$	$P(\varnothing) = 0$
6	9	$S = \{HH, HT, TH, HH\}$	$S = \{HH, HT, TH, TT\}$
7	10	$F_X(x) = 1 - p$ if $x \le x < 1$	$1 - p$ if $0 \le x < 1$
8	11	$E(X) = \sum x_i P(A_i)$ written $\sum x_i E(I_{A_i})$ with $\{A_i\}$ and $\{B_j\}$ called "independent partitions"	partitions of S; independence is not needed for linearity
9	13	Hölder stated as $E\|XY\|^r \le E^{1/r}\|X\|^r E^{1/s}\|Y\|^s$	$E\|XY\| \le \left(E\|X\|^r\right)^{1/r}\left(E\|Y\|^s\right)^{1/s}$
10	12–14	Young's inequality derived from $Q(p) = \frac{p^r}{r} + \frac{p^{-s}}{s}$ with the substitution left implicit	derivation shown, giving $ab \le \frac{a^r}{r} + \frac{b^s}{s}$
11	15	$g(a)P(A) \le E[g(x)] \le \sup g(x)$	$g(a)P(A) \le E[g(X)] \le \sup g(x)\,P(A) + g(a)$
12	15	$\frac{Eg(x) - g(a)}{\sup g(x)} \le p[(x) > a] \le \frac{Eg(x)}{g(a)}$	$\frac{E[g(X)] - g(a)}{\sup g(x)} \le P[g(X) > a] \le \frac{E[g(X)]}{g(a)}$
13	16	Chebyshev with the ϵ omitted throughout	$\epsilon > 0$ restored
14	16	"put X an X_n"	$X = X_n$ in Markov's inequality
15	21	$X_n \ge \ \Rightarrow E(X_n) \ge E(Y_n)$	$X_n \ge Y_n \Rightarrow E(X_n) \ge E(Y_n)$
16	22	$E(X) \le \lim E(X_n) \le \bar{\lim} E(X_n) \le E(\lim E(X_n))$	$E(X) \le \liminf E(X_n) \le \limsup E(X_n) \le E(X)$
17	26	$Q_Y(t) = Q_X(bt)$ with the factor e^{iat} detached	$Q_Y(t) = Q_X(bt)\,e^{iat}$
18	26–27	$\mathrm{Re}(1 - Q(t)) \le \frac{1}{4}(1 - \mathrm{Re}\,Q(2t))$	$\mathrm{Re}(1 - Q(t)) \ge \frac{1}{4}\mathrm{Re}(1 - Q(2t))$

#	Source page	As printed	As corrected
19	29–30	$\mathcal{A}_n^* = \sigma\left[\cup_{i=1}^{\infty} X_i^{-1}B\right]$, $D_n = \sigma\left[\cup_{i=1}^{\infty} X_i^{-1}B\right]$ with the same index	$D_n = \sigma\left[\cup_{i \geq n} X_i^{-1}\mathcal{B}\right]$, so that $D_n \supset D_{n+1}$
20	31	$P\left(\lim B_n\right) = \lim_r \lim_s \left(1 - P\left(\cap_{k=r}^{s} B_k^c\right)\right)$ with B_k^c inside the intersection	$\lim_r \lim_s \left(1 - \prod_{k=r}^{s} \left(1 - P\left(B_k\right)\right)\right)$, using independence
21	32	$P\left[\lvert S_n - np\rvert/n > \epsilon\right] \leq npq/n^2\epsilon^2$	$P\left[\left\lvert S_n/n - p\right\rvert > \epsilon\right] \leq \frac{npq}{n^2\epsilon^2}$
22	14	"JENSENS IN-EQALITY: $FE(x) \leq EF(x)$"	$f\left(E(X)\right) \leq E\left(f(X)\right)$
23	24	$P\left[\lvert A_n\rvert > \epsilon\right] = P\left[\lvert a\rvert\lvert X_n\rvert > \epsilon\right]$	$P\left[\lvert aX_n\rvert > \epsilon\right] = P\left[\lvert X_n\rvert > \epsilon/\lvert a\rvert\right]$

APPENDIX B — SOURCE PAGE MAP

The source document has 33 pages; all are represented above.

Source pages	Content
1	Chapter title; §1.1 definitions (set, null set, universal set, class, subset)
2	equality, superset, disjoint sets, union, intersection, difference, complement, power set
3	§1.2 sequences and limits; monotone increasing/decreasing; Examples 1 and 2; §1.3 fields
4	field rules; §1.3.1 DeMorgan's rule and its proof; Results 1 and 2; minimal field
5	σ-field; Borel field; monotone field; theorem (field $\Leftrightarrow$ monotone field) with proof; §1.3.2 functions
6	one-one/one-many/many-one; real-valued and set functions; inverse functions; measurable and Borel functions; measurable space; measures
7	finite and σ-finite measures; indicator functions (1.1); simple functions (1.2); theorem on simple functions
8	proof of the simple-function theorem; partitions; §1.3.3 probability; experiment, sample space, outcome
9	§1.4 classical, statistical and axiomatic approaches; Table 1.1; random variables
10	distribution functions; example; theorem on distribution functions and proof
11	$F_X(\pm\infty)$; §1.4.1 expectation (1.3); linearity
12	linearity (continued); scale preserving; non-negativity; §1.5 moments; Lemma on $E\|X\|^s$; C_r-inequality
13	C_r-inequality for $r \geq 1$; Hölder's inequality with proof
14	Minkowski's inequality; convex functions; Jensen's inequality
15	Basic inequality: statement, proof, (1.4)–(1.7)
16	conclusion of the basic inequality; Markov and Chebyshev; §1.5.1 convergence in probability
17	Markov-based sufficiency theorem (1.8); the ratio criterion (1.9)–(1.10); §1.5.2 almost sure convergence; §1.5.3 convergence in distribution
18	convergence in distribution implies nothing weaker; proof that convergence in probability gives convergence in distribution
19	(1.11)–(1.13); §1.5.4 convergence in the rth mean, cases $r \leq 1$ and $r \geq 1$
20	case $r \geq 1$ continued; §1.5.5 monotone convergence theorem; §1.6 Fatou's theorem, parts 1 and 2
21	Fatou's theorem continued; §1.7 dominated convergence theorem; §1.7.1 product space
22	product spaces, sections; §1.7.2 mutual convergence; §1.7.3 closure under arithmetic operations
23	proofs of closure under addition and multiplication
24	continuation of the multiplication proof; quotient
25	§1.8 Fubini's theorem; §1.9 characteristic functions; §1.9.1 properties
26	continuity proof; boundedness; linear combination; the lemma on Re $\big(1 - Q(t)\big)$
27	§1.9.2 inversion theorem; §1.9.3 independence
28	null and sure events; §1.9.4–§1.9.5 independent classes; §1.9.6 independent random variables
29	§1.9.7 induced σ-field; §1.9.8 tail σ-field; §1.10 Kolmogorov 0–1 law
30	proof of the 0–1 law; §1.10.1 Borel–Cantelli lemma, part 1
31	Borel–Cantelli part 2; §1.10.2 equivalent sequences; §1.10.3 convergence in probability of S_n
32	conclusion of §1.10.3; §1.10.4 stability; §1.11 Bernoulli weak law

Source pages	Content
33	§1.11.1 central limit theorem; §1.11.2 Lindeberg–Lévy; §1.11.3 Liapounov